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Asphaltene Densities and Solubility Parameter Distributions: Impact on Asphaltene Gradients

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ABSTRACT: Analysis of spatial gradients of the concentration of asphaltenes in reservoir crude oils has become an important tool for oilfield reservoir characterization. Modeling these gradients requires knowledge of asphaltene properties. For example, the commonly employed Flory−Huggins−Zuo (FHZ) equation requires the asphaltene particle size, solubility parameter, and density. Asphaltenes from various sources, particularly including sources beyond conventional crude oil, can have widely varying compositions such as the H/C ratio. Here, we measured the solubility parameters and densities of five asphaltenes that span a large range of H/C ratios, including asphaltenes from coal, petroleum, and shale, and then performed a sensitivity analysis to examine how that range of compositions impacts the magnitude of asphaltene gradients as predicted by the FHZ equation. Two case studies are included in the sensitivity analysis, one in which the asphaltene gradient is driven primarily by gravity, and another in which the asphaltene gradient is driven primarily by solubility. The results shows that the sensitivity of the modeled asphaltene gradients to the measured range of asphaltene solubility parameters and densities is low such that, for both case studies, varying these parameters across the entire measured range does not impact the assignment of the state of asphaltene aggregation determined by the fitted asphaltene particle size. These results suggest that asphaltene gradients can be modeled using the FHZ equation with default values for the asphaltene solubility parameter and density, and local calibration of those parameters will negligibly impact the analysis.

■ INTRODUCTION

Reservoir crude oils consist of dissolved gases, liquids, and dissolved solids (the asphaltenes). It is typically the case that the abundances of these different components are not constant across an entire oil reservoir. Instead, spatial gradients in properties such as the gas-oil ratio (GOR) and the asphaltene (defined as the fraction that is insoluble in aliphatic solvent such as heptane but soluble in aromatic solvent such as toluene) content are typically observed in reservoirs. Analysis of these spatial gradients has become an important industrial process to understand the geologic structure and processes relevant to petroleum reservoirs.[1,2]

Gradients in oil composition are created by numerous forces. One force is gravity, which acts to concentrate relatively low density components such as methane at the top of the reservoir and relatively high density components such as asphaltenes at the bottom of the reservoir. Because methane and asphaltenes are immiscible, solubility interactions create an energetic penalty if high concentrations of methane and asphaltenes occur simultaneously, resulting in the exaggeration of the magnitude of gradients created by gravity. Other forces such as entropy also influence asphaltene gradients, but their effect is typically relatively small.[3]

With sufficient knowledge of the properties of the different fractions of crude oil, the form of these gradients can be predicted. Properties of the volatile fraction of petroleum are well understood owing to sophisticated analyses typically involving gas chromatography. The development of solid-state analyses is extending that understanding into the nonvolatile fraction, particularly the asphaltenes.[4] After considerable experimentation, some bulk properties of petroleum asphaltenes, including their molecular and colloidal sizes, have now been resolved and described by the Yen−Mullins model.[5]

Gradients in crude oil composition can be predicted using algorithms based on equation of state (EoS) models. For example, versions of the cubic EoS model gas−liquid interactions and can predict GOR gradients based on composition typically measured by gas chromatography.[6] Asphaltene phase behavior has been modeled by Flory−Huggins theory developed originally to treat polymer solutions.[7] The Flory−Huggins−Zuo (FHZ) theory additionally incorporates the gravity effect, resulting in models of the asphaltene gradients based on asphaltene composition as described by the Yen−Mullins model.[8,9]

Application of these models for oilfield formation evaluation is facilitated by logging tools capable of performing downhole fluid analysis (DFA).[1] These tools provide in situ measurements of many reservoir fluid properties during or immediately after drilling the well. Physical properties such as density and

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viscosity can be measured within single wells and across oilfields in multiple wells using various DFA tools. Particularly relevant to gradient analysis, DFA tools measure the concentration of dissolved gases, liquids, and asphaltenes using optical spectroscopy.

The combination of the Yen−Mullins model of asphaltene structure, FHZ model of asphaltene gradients, and DFA measurement of asphaltene gradients has been used for numerous investigations on the geologic structure of petroleum reservoirs as well as the history of geologic processes that have occurred in those reservoirs. One aspect of geologic structure particularly relevant to oilfield formation evaluation is the ability of reservoir fluids to flow from one section of a formation to another, often referred to as reservoir connectivity.[2] The FHZ model predicts asphaltene gradients at equilibrium as would result if the asphaltenes could migrate throughout an entire connected reservoir over long periods of time; therefore, the connectivity of a reservoir can be investigated by comparing measured asphaltene gradients to those predicted by the FHZ model.[9−12] Additionally, the model has been applied to address reservoirs whose structures are impacted by fault block migration[13] as well as reservoirs subjected to various processes, including tar mat formation,[14−18] biodegradation,[19] and water washing.[20] The FHZ method has also been adapted to handle nonequilibrium situations for analysis of reservoirs where dynamic processes are active or recently completed.[19,21−23]

Three properties of asphaltenes are involved in gradient modeling with the FHZ: the particle size, solubility parameter, and density. Asphaltene particle size is described by the Yen−Mullins model, which states that asphaltenes are found as either free molecules, nanoaggregates consisting of approximately six to eight molecules, or clusters consisting of approximately eight nanoaggregates.[5,24,25] After considerable controversy,[26,27] advances in mass spectrometry, particularly laser desorption laser ionization mass spectrometry (L^2MS), have converged on a number-average petroleum asphaltene molecule weight of approximately 600 g/mol.[28−31] Measurements of the sizes of the nanoaggregates and clusters have been performed using techniques such as surface-assisted laser desorption ionization (SALDI) mass spectrometry,[31,32] conductivity,[33−35] NMR,[36,37] and small-angle X-ray and neutron scattering.[38−41]

The Hildebrand solubility parameter is a single chemical parameter that describes intermolecular interactions. Studies of the projection of the asphaltene Hildebrand solubility parameter into the Hansen components show that the polarizability term is dominant,[42] thereby enhancing the rationale for this simple approach for crude oils.[43] Solubility parameters were developed originally for liquid state applications. However, over the years, they have been successfully used for many other applications, including but not limited to polymers, dyes, emulsions, etc.[44]

Determining the solubility parameter of a low molecular weight liquids is a straightforward task. It requires knowing only the vaporization energy and liquid molar volume.[45] However, for solid materials, the solubility parameter concept cannot be applied directly because the reference state is the liquid substance. For those cases, it is necessary to use a hypothetical liquid subcooled below the melting point of the substance as a reference.[44,46] Because of this limitation, a series of different methods have been developed to determine solubility parameters of solids. As a result, solubility parameters of solids are dependent on the method used to determine them.[44] The most popular method used to evaluate asphaltene solubility parameters is a Heithaus titration based on the regular solution model.[47] Other methods used to determine asphaltene solubility parameters include solubility testing in different solvents[48] as well as molecular modeling[49] and IR-NIR multivariable analysis.[50]

Complementing previous measurements of the average solubility parameters, it is also possible to measure the distribution of solubility parameters of mixtures such as asphaltenes. This so-called "asphaltene solubility profile" can be used to determine the colloidal stability of crude oils and correlates remarkably well with common stability measurements.[51,52] A study of samples from different origins indicated that different solubility distribution patterns can be correlated with precipitation tendencies of crude oils.[51] Even though the solubility distribution is originally obtained as a function of time in the experiments, it can be converted to a distribution as a function of solubility parameter.[53] Two procedures are possible to perform this conversion. One procedure uses the Hildebrand−Scatchard expression,[44] which shows that dissolution of a solute in a solvent is favored when the solubility parameters of the solute and solvent match. The second procedure is based on the calibration of the solubility profile time scale using asphaltene fractions with different known parameters calculated from experimental densities and the Third Rule that correlates density with solubility parameter for hydrocarbon molecules.[54] Regarding solubility parameter values, this last procedure gives values that are within the range reported by other techniques,[48−50] whereas the Hildebrand−Scatchard expression yields lower values.[53]

Previous investigations have shown a great diversity in the composition of asphaltenes, particularly asphaltenes extracted from different sources. A collection of petroleum asphaltenes from a large number of conventional reservoirs span a range of H/C ratios from 0.92 to 1.56, with an average H/C of 1.13 and a standard deviation of 0.11.[55] Petroleum asphaltenes from the Northern Viking Graben area in the Norwegian North Sea were found to range from 0.9 to 1.5 in H/C ratio.[56] Asphaltenes from regions sourced by the immature Type I Green River Shale have relatively high H/C values ranging from 1.30 to 1.67.[57−59] Asphaltenes from coal liquefaction typically have lower H/C ratios (approximately 0.8) and smaller molecular sizes.[29,60,61] Asphaltenes formed at a range of maturities have large and systematic variations in their H/C ratio, molecular weight, sulfur speciation, oxygenation, aromaticity, degree of condensation, and chain length, with variation in the molecular weight and heteroatom-containing functional groups occurring primarily at low maturities, while variation in the carbon backbone occurs mainly at higher maturities.[62]

Given the diversity in composition of asphaltenes from various sources, it might be expected that asphaltenes also present variability in their properties that are relevant for asphaltene gradient modeling, i.e., their solubility parameter and density. Some differences in solubility parameters have been observed for asphaltenes from oil sands bitumens, oils, asphalts, and deposits, and those differences have been related to the composition of the asphaltenes.[63,64] Here, we present measured values of the solubility parameter distribution and the skeletal density for three classes of asphaltenes: asphaltenes from coal liquefaction (relatively low H/C), asphaltenes from petroleum (moderate H/C), and asphaltenes from immature source rock (shale asphaltene, high H/C). Subsequently, two case studies, one with asphaltene gradient driven primarily by

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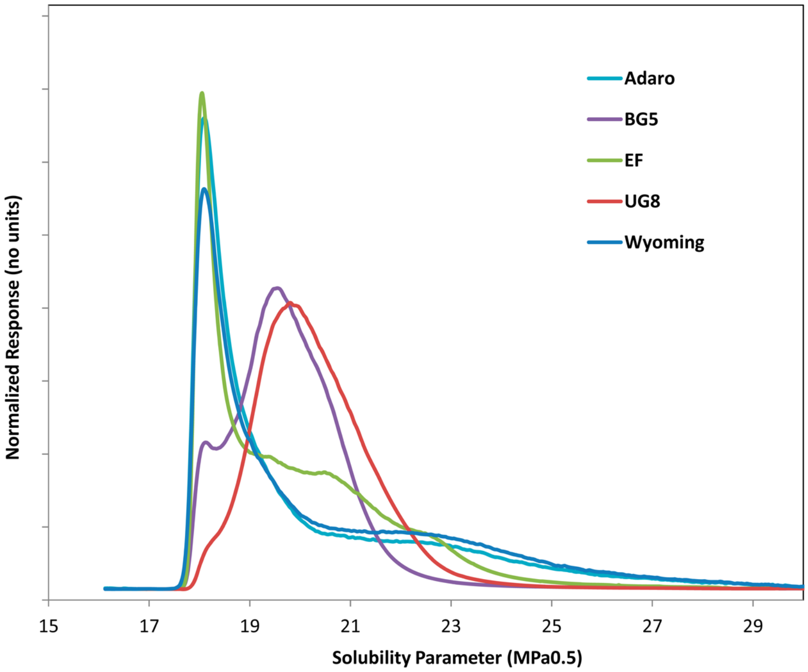

Figure 1. Asphaltene solubility profiles.

gravity, and another with asphaltene gradient driven primarily by solubility, were considered to conduct a sensitivity analysis of the magnitude of the variation in predicted asphaltene gradients when the asphaltene solubility parameter and density are varied over the measured range. From this sensitivity analysis, conclusions are drawn regarding the uncertainty in asphaltene gradient prediction in common situations where the asphaltene solubility parameters and density are uncertain.

■ EXPERIMENTAL SECTION

Samples. Measurements were performed on five asphaltene samples: two petroleum asphaltenes, one shale asphaltene, and two coal asphaltenes. The petroleum asphaltenes (UG8 and BG5) were extracted from Kuwaiti oil and have been analyzed previously by other methods.[29,61] Asphaltenes were extracted from the oil by having the oil diluted 1/40 in *n*-heptane, stirred overnight, and then filtered. The collected solids were then purified by washing for several days with *n*-heptane in a Soxhlet extractor.

The shale asphaltene (Eagle Ford) was obtained from an immature outcrop (% VRo = 0.65) of the organic-rich lower Eagle Ford formation in road cuts near Del Rio, Texas.[65] Kerogen in this formation is type II. The total organic content as well as U, Th, and K signatures of the outcrops matched those of a nearby well drilled in areas with a thick cover of Austin Chalk to ensure that there was no contact with meteoric water, indicating insignificant weathering of the outcrop samples. Soluble organic matter (bitumen) was extracted from the outcrop by Soxhlet extraction using dichloromethane. Asphaltenes were then extracted and purified from the bitumen by *n*-heptane precipitation as described above.

The coal asphaltenes (Adaro, Wyoming) were extracted from the residue of a coal liquefaction process as described previously.[60] The samples originated from Indonesia (Adaro) and the United States (Wyoming). The asphaltenes were extracted from the residue in the manner described above, except *n*-hexane was used as the solvent.

Skeletal Density. The skeletal density of the asphaltene samples was measured by helium pycnometry using a Micromeritics AccuPyc II 1340 gas pycnometer. The pycnometer measures the volume of the solid material (including any disconnected pores) by displacement of helium gas. The sample was also weighed, and the density was obtained by dividing the measured mass by the measured volume. The pycnometer was calibrated using NIST calibration spheres. The error of the measurement was approximately ±0.03%.

Solubility Profile Analysis. Solubility profile analyses were carried out for all the asphaltenes using the following procedure: solutions of the samples in methylene chloride (0.0100 g in 10 mL) were prepared and injected (80 μL) in a column packed with an inert material using *n*-heptane as the mobile phase. This solvent induces the precipitation of asphaltenes and, as a consequence, their retention in the column. The first eluted fraction from the column is the maltenes, which are soluble in *n*-heptane. After all of this fraction has eluted, the mobile phase is changed gradually from pure *n*-heptane to 90/10 methylene chloride/methanol and then to 100% methanol. Asphaltenes were quantified using an evaporative light scattering detector (ELSD). The HPLC system consisted of an HP Series 1100 chromatograph and an Alltech ELSD 2000 detector. A detailed account of the technique is presented elsewhere.[51,52] Usually, heptane extracted asphaltenes from virgin materials do not produce maltene peaks during the test.[51]

The procedure redissolves the asphaltenes gradually from the EDA (low solubility parameter: easy to dissolve asphaltenes) to the DDA (high solubility parameter: difficult to dissolve asphaltenes). On the basis of the ELSD detector, a curve is generated that is related to the solubility properties of the asphaltenes as a function of time and can be quantified to reflect the tendency of the sample toward asphaltene precipitation. Two main variables can be calculated using this procedure: T_{av}, the average time of elution of the asphaltenes; and ΔPS, which reflects the tendency of asphaltenes toward precipitation and is usually applied to crude oils/products. In the present work, T_{av} was used in the analyses of the asphaltenes as it is proportional to the average solubility parameter of the asphaltenes. On the basis of this, a correlation previously established[53] for asphaltenes was used in the calculations of the asphaltene solubility parameters.

Heptane asphaltene extracted from virgin crude oils or residues does not show a maltene peak during the test.[66] This is not the case for the asphaltenes extracted from the two coal-derived samples (Adaro and Wyoming), which were extracted using *n*-hexane.[60] Because of the large separation between the maltene and asphaltene peak, the maltenes in the coal samples do not affect the measured asphaltene solubility parameters.[66]

Flory–Huggins–Zuo Equation of State. Asphaltene gradients were predicted based on the FHZ EoS.[8,9] The FHZ equation describes asphaltene gradients according to

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Table 1. Average Properties of Asphaltenes

type	name	H/C ratio	skeletal density (g/cm^3)	solubility parameter (MPa$^{0.5}$) mean	median	mode	standard deviation
shale	Eagle Ford	1.39	1.15	19.9	19.4	18.1	1.9
petroleum	BG5	1.18	1.17	19.8	19.7	19.5	1.2
petroleum	UG8	1.05	1.19	20.3	20.2	19.7	1.3
coal	Adaro	0.85	1.21	20.1	19.0	18.1	2.6
coal	Wyoming	0.81	1.26	20.4	19.3	18.1	2.6

[a]Typical measurement errors are 0.05 for H/C ratio and 0.03 g/cm^3 for skeletal density.

$$\frac{\varphi_a(h_2)}{\varphi_a(h_1)} = \exp\left\{\frac{v_a g(\rho - \rho_a)(h_2 - h_1)}{RT} + \frac{v_a}{RT}[(\delta_a - \delta)^2_{h_1} - (\delta_a - \delta)^2_{h_2}] + \left[\left(\frac{v_a}{v}\right)_{h_2} - \left(\frac{v_a}{v}\right)_{h_1}\right]\right\}$$

The variables ϕ, R, v, δ, T, g, ρ, and h are the volume fraction, universal gas constant, molar volume, solubility parameter, temperature, earth's gravitational acceleration, density, and depth, respectively. Subscript a denotes the properties of asphaltenes; subscripts h_1 and h_2 stand for the properties at depths h_1 and h_2, respectively. This equation considers the three forces driving asphaltene content gradients of gravity, solubility, and entropy in the three terms, respectively.

Here, gradients were predicted for reservoirs with known values of $\phi_a(h)$, T, δ, and ρ using the range of measured values of δ_a and ρ_a determined here. v_a is a fitting parameter and should conform to one of the three allowed asphaltene particle sizes described by the Yen–Mullins model for connected and equilibrated reservoirs. A sensitivity analysis is performed by varying δ_a and ρ_a over the measured range while fitting the measured $\phi_a(h)$ to the FHZ EoS and observing the resulting variation in v_a.

RESULTS

Figure 1 shows the solubility parameter distributions obtained for the asphaltenes. For comparison proposes, all curves have been normalized so that the areas under the curves are equal to one. The distributions for the two coal asphaltenes show peaks at a solubility parameter lower than that found for asphaltenes from traditional crude oils. The latter samples show the standard behavior for this type of asphaltene.[51] This low solubility parameter peak is associated with easy-to-dissolve asphaltenes.[51] However, the solubility profiles for the coal asphaltenes show a tail that extends into solubility parameters larger than what is observed for the solubility profiles of the petroleum asphaltenes.

Eagle Ford asphaltenes, extracted from tight oil shale, also exhibit behavior different from that observed for asphaltenes from traditional crude oils. In a similar fashion to coal asphaltenes, they elute at retention times remarkably lower than those of the standard heptane asphaltenes, showing the large peak corresponding with easy-to-dissolve asphaltenes. This phenomenon can be noticed by comparing the profiles of Eagle Ford asphaltenes with BG5 and UG8 petroleum asphaltenes.

Table 1 reports the average properties of the asphaltene samples. Samples are listed in order of increasing atomic H/C ratio. A wide range of H/C ratios is observed, with the shale asphaltene having relatively high H/C, the petroleum asphaltenes having moderate H/C, and the coal asphaltenes having relatively low H/C. High H/C generally correlates with low aromaticity for carbonaceous materials.[67,68] Average solubility parameters are also listed. Despite the diversity of asphaltene H/C ratios and significant differences in the width of the solubility parameter distribution as measured by the solubility profile, the average solubility parameters of all the asphaltenes are within a small range (19.8–20.4 MPa$^{0.5}$). These values are within the expected range previously reported for asphaltenes using different techniques.[50] Additionally, several other statistical parameters describing the solubility profiles are presented. The petroleum asphaltenes are found to be relatively monodisperse, whereas the shale asphaltenes and particularly the coal asphaltenes are more polydisperse.

The measured asphaltene skeletal densities lie in a range (1.15–1.26 g/cm^3) centered around the value of 1.2 g/cm^3 commonly used for asphaltene gradient analysis.[9,69,70] Figure 2 plots the asphaltene average solubility parameter and density as a function of H/C ratio. Both parameters generally decrease with increasing H/C, although there is some scatter in the relationship with solubility parameter. A similar decrease in solubility parameter with increasing H/C is commonly observed in carbonaceous materials.[64] The inverse relationship between density and H/C ratio is similar to what has been observed previously for kerogen, where decreasing H/C ratio[71] and increasing density[72] are both correlated with increasing maturity.

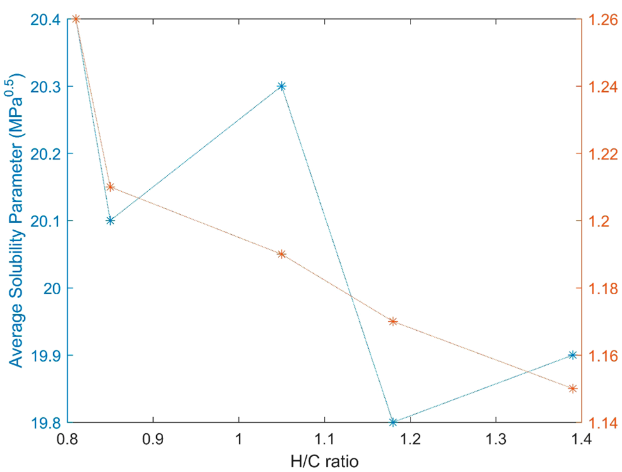

Figure 2. Asphaltene solubility parameter (left scale, in blue) and density (right scale, in red) generally decrease with increasing H/C ratio.

DISCUSSION

Using the ranges of asphaltene solubility parameter and density measured above, the impact of those properties on asphaltene gradients is examined below. In particular, sensitivity analyses

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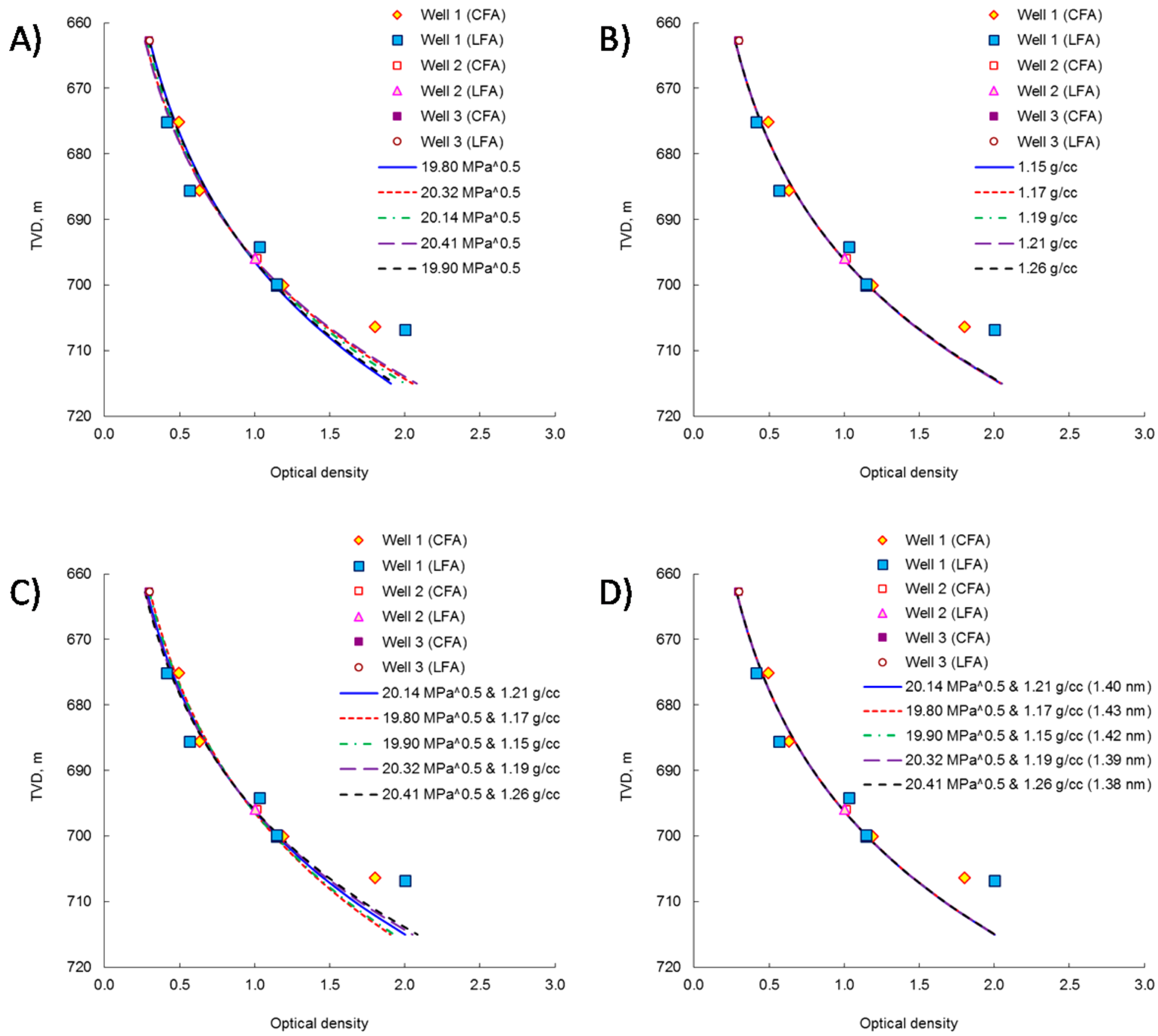

Figure 3. FHZ fits for the asphaltene gradient driven primarily by solubility. TVD are referenced to an arbitrary location because only the relative depths are significant for this analysis. Data points are labeled according to the DFA logging tool used to make the measurement.[1] Panel A: particle size and density held fixed (1.4 nm and 1.2 g/cm^3) while solubility parameter is varied. Panel B: particle size and solubility parameter held fixed (1.4 nm and 20.314 MPa$^{0.5}$) while density is varied. Panel C: particle size held fixed (1.4 nm) while solubility parameter and density are varied simultaneously. Panel D: solubility parameter and density held fixed (at the measured values) while particle size is optimized in the fit.

are performed wherein measured asphaltene gradients are fit with the FHZ, while the values of solubility parameter and density used in the fitting are tuned over the measured ranges of average solubility parameters and skeletal densities. Asphaltene gradients are typically driven by solubility and/or gravity because the entropy term is generally small.[3] Thus, two sensitivity analyses are performed: one for a reservoir in which the asphaltene gradient is dominated by solubility, and another where the asphaltene gradient is dominated by gravity.

In the first case study, a large reservoir in the North Sea contains a relatively light fluid in which the asphaltene gradient is driven primarily by solubility.[73,74] Figure 3 presents the observed gradient in asphaltene content, measured in multiple wells by downhole fluid analysis tools.[73] The graphs plot the optical density (proportional to the asphaltene content) against the true vertical depths (TVD) where the oils were acquired. In addition to the measured gradients, the figure also shows FHZ fits using different values of the parameters. Typically all FHZ parameters are fixed during the fitting except the asphaltene particle size, which is used as a fitting parameter. In the sensitivity analysis presented here, the asphaltene solubility parameter and density are varied across the measured range. If the default asphaltene solubility parameter (20.314 MPa$^{0.5}$) and asphaltene density (1.2 g/cm^3) are used, the FHZ fits these data to an asphaltene particle size of 1.4 nm. This particle size suggests that asphaltenes in this reservoir exist as isolated molecules, as expected for light oils such as those that occur here.[3]

Figure 3a shows the FHZ fits where the particle size and density are held fixed while the asphaltene solubility parameter is varied. Different solubility parameters result in slightly different curves, although the difference is small, and all curves fit the measured data approximately equally well. Figure 3b shows the FHZ fits where the particle size and solubility parameter are held fixed while the asphaltene density is varied. The curves essentially all overlay, which is expected because the

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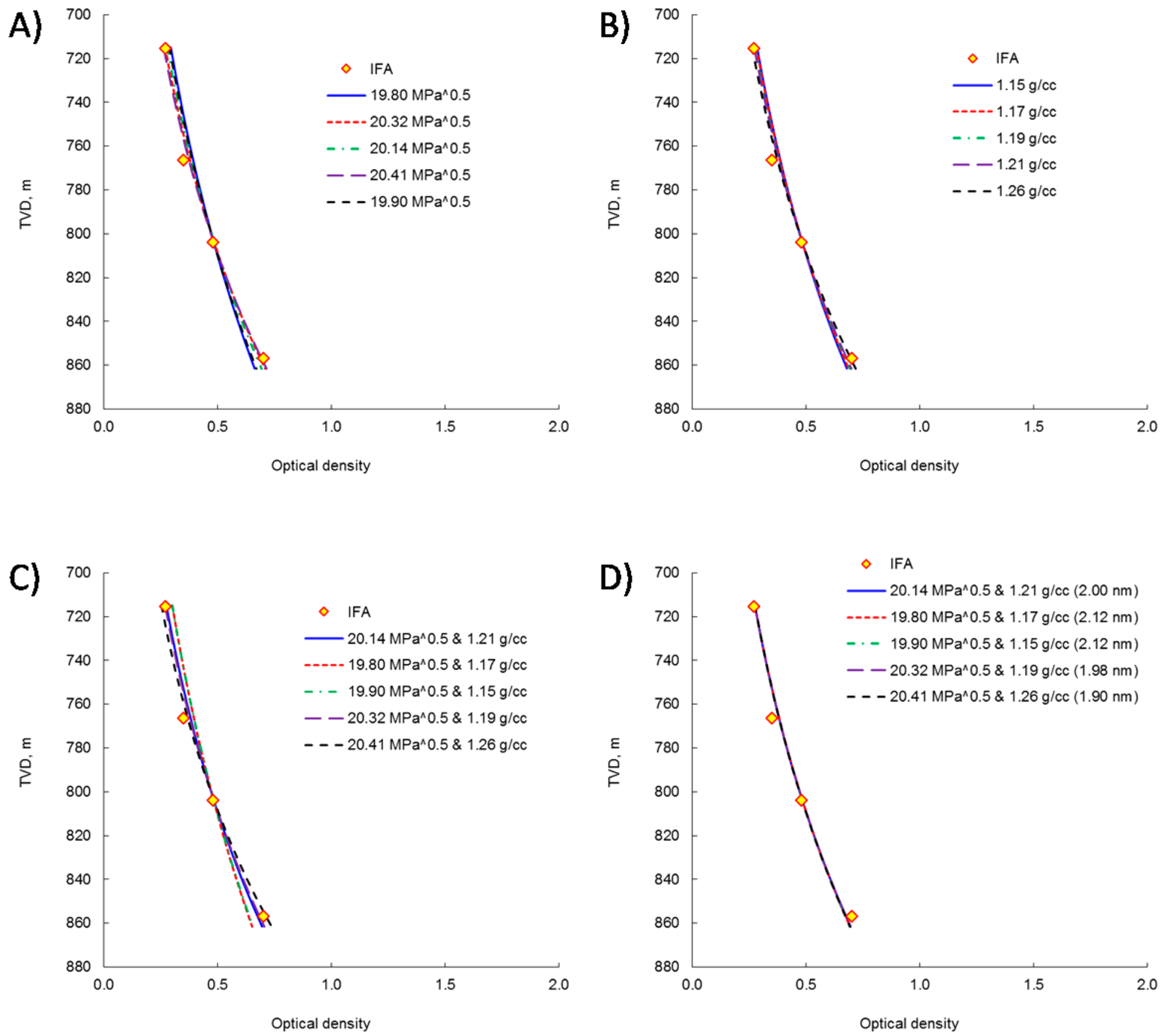

Figure 4. FHZ fits for the asphaltene gradient driven primarily by gravity. TVD are referenced to an arbitrary location because only the relative depths are significant for this analysis. Data points are labeled according to the DFA logging tool used to make the measurement.[1] Panel A: particle size and density held fixed (2.0 nm and 1.2 g/cm^3) while solubility parameter is varied. Panel B: particle size and solubility parameter held fixed (2.0 nm and 20.2 MPa$^{0.5}$) while density is varied. Panel C: particle size held fixed (2.0 nm) while solubility parameter and density are varied simultaneously. Panel D: solubility parameter and density held fixed (at the measured values) while particle size is optimized in the fit.

gradient in this example is driven by solubility, and the asphaltene density does not appear in the solubility term. Figure 3c shows the FHZ fits where the particle size is held fixed (at 1.4 nm) while the asphaltene solubility parameter and density are both varied simultaneously. The five curves represent the five (solubility parameters, density) pairs resulting from the five asphaltenes measured. These curves are nearly identical to those where only the asphaltene solubility parameter is varied (Figure 3a), consistent with the asphaltene density having a negligible impact in this reservoir.

Figure 3d shows the FHZ fits where the solubility parameter and density pairs are held fixed at the measured values (as in Figure 3c) while the asphaltene particle size is optimized by the fit. Variations in the optimized particle size can compensate for essentially all of the deviation in the predicted asphaltene gradients resulting from differences in solubility parameter and density, and as a result, the five curves nearly overlay. Moreover, the range of the fitted values of the asphaltene particle size is small: 1.38−1.43 nm. In each case, the asphaltene particles would be assigned as isolated molecules (nominally 1.5 nm according to the Yen−Mullins model).[5]

In the second case study, a deepwater field in the Gulf of Mexico contains a black oil with low gas/oil ratio, and the asphaltene gradient here is driven primarily by gravity.[75] Figure 4 presents the FHZ sensitivity analysis for this case. If the default asphaltene solubility parameter and asphaltene density are used, the FHZ fits these data to an asphaltene particle size of 2.0 nm. This particle size suggests that asphaltenes in this reservoir exist as nanoaggregates, as expected for low gas/oil ratio black oils such as those that occur here.[3]

Figure 4a shows the FHZ fits where the particle size and density are held fixed while the asphaltene solubility parameter is varied. Different solubility parameters result in slightly different curves, although the difference is small, and all curves fit the measured data approximately equally well. Some variation between the curves is apparent, demonstrating that the solubility interaction contributes even in this reservoir where the gradient is driven primarily by gravity. Figure 4b

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shows the FHZ fits where the particle size and solubility parameter are held fixed while the asphaltene density is varied. Some variation between the curves is apparent here as well, although again, the difference is small, and all curves fit the measured data approximately equally well. A greater difference is observed here relative to that in the reservoir dominated by solubility (Figure 3b), illustrating the greater impact of gravity in this reservoir.

Figure 4c shows the FHZ fits where the particle size is held fixed (at 2.0 nm) while the asphaltene solubility parameter and density are both varied simultaneously. The variation between the curves is again small, although larger than that in Figures 4a or b. Because the asphaltene solubility parameter and asphaltene density are covariant (both are above their averages for relatively low H/C asphaltenes and below their averages for relatively high H/C asphaltenes, see Figure 2), the resulting variations in both parameters amplify the deviations in the predicted asphaltene gradients. Nevertheless, the variations between the curves are still rather small. Expressed more quantitatively, Figure 4d shows the FHZ fits where the solubility parameter and density pairs are held fixed while the asphaltene particle size is optimized by the fit. Variations in the fitted particle size again compensate for essentially all of the deviation in the predicted asphaltene gradients resulting from differences in solubility parameter and density, and again, the five curves nearly overlay. The range of the fitted values of the asphaltene particle size is small: 1.90−2.12 nm. In each case, the asphaltene particles would be assigned as nanoaggregates (nominally 2.0 nm according to the Yen−Mullins model).[5]

■ SUMMARY AND CONCLUSION

Modeling spatial gradients in asphaltene content of reservoir fluids requires knowledge of three parameters describing asphaltenes: their particle size, solubility parameter, and density. Asphaltenes from various sources, particularly including sources beyond conventional petroleum reservoirs, can have significantly different compositions such as different H/C ratios. However, H/C ratio does not directly impact asphaltene gradient modeling. To understand the effect of compositional variability on the factors directly affecting asphaltene gradient modeling, asphaltenes from three different sources (coal asphaltenes with H/C ratio around 0.8, petroleum asphaltenes with H/C ratio near 1.1, and shale asphaltenes with H/C ratio around 1.4) were analyzed for their skeletal density and solubility profile. The measured densities of these asphaltenes were found in the range of 1.15−1.26 g/cm^3, and the measured average solubility parameters of these asphaltenes were found in the range of 19.8−20.4 MPa$^{0.5}$. To test the significance of these ranges, sensitivity analyses were performed, where two case studies with measured asphaltene gradients, one driven primarily by gravity and the other driven primarily by solubility, were fitted to the FHZ equation using the range of measured asphaltene densities and solubility parameters. It was found the gradients predicted using this range of asphaltene density and solubility parameters were nearly identical: all gradients fit the measured data nearly equally well, and variations in the asphaltene density and solubility parameter can be compensated for by variations in the fitted asphaltene particle size that lie within the normal range of sizes for a particular asphaltene aggregate, as described by the Yen−Mullins model.

These results lead to the conclusion that the natural range of asphaltene densities and solubility parameters is sufficiently small and has a negligible effect on asphaltene gradient modeling. Even though other properties describing asphaltenes can vary greatly, including the width of the solubility parameter distribution, which impacts asphaltene flow assurance, the average solubility parameter and density vary over small ranges that do not significantly impact gradients. Therefore, asphaltene gradients can be modeled using the FHZ equation with default values of asphaltene solubility parameter and density, and local calibration of those parameters will negligibly impact the analysis.

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Notes

The authors declare no competing financial interest.

■ REFERENCES

(1) Mullins, O. C. *The Physics of Reservoir Fluids: Discovery Through Downhole Fluid Analysis*; Schlumberger: Sugar Land, TX, 2008; p 184.

(2) Mullins, O. C.; Zuo, J. Y.; Wang, K.; Hammond, P. S.; de Santo, I.; Dumont, H.; Mishra, V.; Chen, L.; Pomerantz, A. E.; Dong, C.; Elshahawi, H.; Seifert, D. J. The Dynamics of Reservoir Fluids and Their Substantial Systematic Variations. *Petrophysics* **2014**, *55*, 96−112.

(3) Freed, D. E.; Mullins, O. C.; Zuo, J. Y. Heuristics for Equilibrium Distributions of Asphaltenes in the Presence of GOR Gradients. *Energy Fuels* **2014**, *28*, 4859−4869.

(4) Pomerantz, A. E. Toward Molecule-Specific Geochemistry of Heavy Ends: Application to the Upstream Oil Industry. *Ind. Eng. Chem. Res.* **2016**, *55*, 4403−4414.

(5) Mullins, O. C.; Sabbah, H.; Eyssautier, J.; Pomerantz, A. E.; Barré, L.; Andrews, A. B.; Ruiz-Morales, Y.; Mostowfi, F.; McFarlane, R.; Goual, L.; Lepkowicz, R.; Cooper, T.; Orbulescu, J.; Leblanc, R. M.; Edwards, J.; Zare, R. N. Advances in Asphaltene Science and the Yen−Mullins Model. *Energy Fuels* **2012**, *26*, 3986−4003.

(6) Peng, D.-Y.; Robinson, D. B. A New Two-Constant Equation of State. *Ind. Eng. Chem. Fundam.* **1976**, *15*, 59−64.

(7) Buckley, J. S.; Wang, J.; Creek, J. L. Solubility of the Least-Soluble Asphaltenes. In *Asphaltenes, Heavy Oils, and Petroleomics*; Mullins, O. C., Sheu, E. Y., Hammami, A., Marshall, A. G., Eds.; Springer: New York, 2007; pp 401−438.

(8) Freed, D. E.; Mullins, O. C.; Zuo, J. Y. Theoretical Treatment of Asphaltene Gradients in the Presence of GOR Gradients. *Energy Fuels* **2010**, *24*, 3942−3949.

(9) Zuo, J. Y.; Mullins, O. C.; Freed, D.; Elshahawi, H.; Dong, C.; Seifert, D. J. Advances in the Flory−Huggins−Zuo Equation of State for Asphaltene Gradients and Formation Evaluation. *Energy Fuels* **2013**, *27*, 1722−1735.

(10) Mullins, O. C.; Betancourt, S. S.; Cribbs, M. E.; Dubost, F. X.; Creek, J. L.; Andrews, A. B.; Venkataramanan, L. The Collodial Structure of Crude Oil and the Structure of Oil Reservoirs. *Energy Fuels* **2007**, *21*, 2785−2794.

(11) Pomerantz, A. E.; Ventura, G. T.; McKenna, A. M.; Cañas, J.; Auman, J.; Koerner, K.; Curry, D.; Nelson, R. K.; Reddy, C. M.; Rodgers, R. P.; Marshall, A. G.; Peters, K. E.; Mullins, O. C. Combining biomarker and bulk compositional gradient analysis to assess reservoir connectivity. *Org. Geochem.* **2010**, *41*, 812−821.

(12) Dong, C.; Petro, D.; Pomerantz, A. E.; Nelson, R. K.; Latifzai, A. S.; Nouvelle, X.; Zuo, J. Y.; Reddy, C. M.; Mullins, O. C. New thermodynamic modeling of reservoir crude oil. *Fuel* **2014**, *117*, 839−850.

(13) Dong, C.; Hows, M. P.; Cornelisse, P. M.; Elshahawi, H. *Fault Block Migrations Inferred from Asphaltene Gradients*; SPWLA 54th Annual Logging Symposium: New Orleans, LA, 2013.

(14) Mullins, O. C.; Seifert, D. J.; Zuo, J. Y.; Zeybek, M. Clusters of Asphaltene Nanoaggregates Observed in Oilfield Reservoirs. *Energy Fuels* **2013**, *27*, 1752−1761.

DOI: 10.1021/acs.energyfuels.6b01794
Energy Fuels 2016, 30, 9132−9140

(15) Pomerantz, A. E.; Seifert, D. J.; Bake, K. D.; Craddock, P. R.; Mullins, O. C.; Kodalen, B. G.; Mitra-Kirtley, S.; Bolin, T. B. Sulfur Chemistry of Asphaltenes from a Highly Compositionally Graded Oil Column. *Energy Fuels* **2013**, *27*, 4604−4608.
(16) Wu, Q.; Seifert, D. J.; Pomerantz, A. E.; Mullins, O. C.; Zare, R. N. Constant Asphaltene Molecular and Nanoaggregate Mass in a Gravitationally Segregated Reservoir. *Energy Fuels* **2014**, *28*, 3010−3015.
(17) Forsythe, J. C.; Pomerantz, A. E.; Seifert, D. J.; Wang, K.; Chen, Y.; Zuo, J. Y.; Nelson, R. K.; Reddy, C. M.; Schimmelmann, A.; Sauer, P.; Peters, K. E.; Mullins, O. C. A Geological Model for the Origin of Fluid Compositional Gradients in a Large Saudi Arabian Oilfield: An Investigation by Two-Dimensional Gas Chromatography (GC × GC) and Asphaltene Chemistry. *Energy Fuels* **2015**, *29*, 5666−5680.
(18) Pomerantz, A. E.; Seifert, D. J.; Qureshi, A.; Zeybek, M.; Mullins, O. C. The Molecular Composition of Asphaltenes in a Highly Compositionally Graded Column. *Petrophysics* **2013**, *54*, 427−238.
(19) Zuo, J. Y.; Jackson, R.; Agarwal, A.; Herold, B.; Kumar, S.; Santo, I. D.; Dumont, H.; Ayan, C.; Beardsell, M.; Mullins, O. C. Diffusion Model Coupled with the Flory−Huggins−Zuo Equation of State and Yen−Mullins Model Accounts for Large Viscosity and Asphaltene Variations in a Reservoir Undergoing Active Biodegradation. *Energy Fuels* **2015**, *29*, 1447−1460.
(20) Bartha, A.; De Nicolais, N.; Sharma, V.; Roy, S. K.; Srivastava, R.; Pomerantz, A. E.; Sanclemente, M.; Perez, W.; Nelson, R. K.; Reddy, C. M.; Gros, J.; Arey, J. S.; Lelijveld, J.; Dubey, S.; Tortella, D.; Hantschel, T.; Peters, K. E.; Mullins, O. C. Combined Petroleum System Modeling and Comprehensive Two-Dimensional Gas Chromatography To Improve Understanding of the Crude Oil Chemistry in the Llanos Basin, Colombia. *Energy Fuels* **2015**, *29*, 4755−4767.
(21) Wang, K.; Zuo, J. Y.; Chen, Y.; Mullins, O. C. The dynamic Flory-Huggins-Zuo equation of state. *Energy* **2015**, *91*, 430−440.
(22) Zuo, J. Y.; Chen, Y.; Pan, S.; Wang, K.; Mullins, O. C. Investigation of density inversion induced by gas charges into oil reservoirs using diffusion equations. *Energy* **2016**, *100*, 199−216.
(23) Wang, K.; Kauerauf, A. I.; Zuo, J. Y.; Chen, Y.; Dong, C.; Elshahawi, H.; Mullins, O. C. Differing Equilibration Times of GOR, Asphaltenes and Biomarkers as Determined by Charge History and Reservoir Fluid Geodynamics. *Petrophysics* **2015**, *56*, 440−456.
(24) Mullins, O. C. The Modified Yen Model. *Energy Fuels* **2010**, *24*, 2179−2207.
(25) Mullins, O. C.; Pomerantz, A. E.; Zuo, J. Y.; Dong, C. Downhole fluid analysis and asphaltene science for petroleum reservoir evaluation. *Annu. Rev. Chem. Biomol. Eng.* **2014**, *5*, 325−45.
(26) Herod, A. A.; Bartle, K. D.; Kandiyoti, R. Characterization of Heavy Hydrocarbons by Chromatographic and Mass Spectrometric Methods: An Overview. *Energy Fuels* **2007**, *21*, 2176−2203.
(27) Mullins, O. C.; Martínez-Haya, B.; Marshall, A. G. Contrasting Perspective on Asphaltene Molecular Weight. This Comment vs the Overview of A. A. Herod, K. D. Bartle, and R. Kandiyoti. *Energy Fuels* **2008**, *22*, 1765−1773.
(28) Pomerantz, A. E.; Hammond, M. R.; Morrow, A. L.; Mullins, O. C.; Zare, R. N. Two-Step Laser Mass Spectrometry of Asphaltenes. *J. Am. Chem. Soc.* **2008**, *130*, 7216−7217.
(29) Pomerantz, A. E.; Hammond, M. R.; Morrow, A. L.; Mullins, O. C.; Zare, R. N. Asphaltene Molecular-Mass Distribution Determined by Two-Step Laser Mass Spectrometry. *Energy Fuels* **2009**, *23*, 1162−1168.
(30) Sabbah, H.; Pomerantz, A. E.; Wagner, M.; Müllen, K.; Zare, R. N. Laser Desorption Single-Photon Ionization of Asphaltenes: Mass Range, Compound Sensitivity, and Matrix Effects. *Energy Fuels* **2012**, *26*, 3521−3526.
(31) Pomerantz, A. E.; Wu, Q.; Mullins, O. C.; Zare, R. N. Laser-Based Mass Spectrometric Assessment of Asphaltene Molecular Weight, Molecular Architecture, and Nanoaggregate Number. *Energy Fuels* **2015**, *29*, 2833−2842.
(32) Wu, Q.; Pomerantz, A. E.; Mullins, O. C.; Zare, R. N. Laser-Based Mass Spectrometric Determination of Aggregation Numbers for Petroleum- and Coal-Derived Asphaltenes. *Energy Fuels* **2014**, *28*, 475−482.
(33) Zeng, H.; Song, Y.-Q.; Johnson, D. L.; Mullins, O. C. Critical Nanoaggregate Concentration of Asphaltenes by Direct-Current (DC) Eletrical Conductivity. *Energy Fuels* **2009**, *23*, 1201−1208.
(34) Goual, L.; Sedghi, M.; Mostowfi, F.; McFarlane, R.; Pomerantz, A. E.; Saraji, S.; Mullins, O. C. Cluster of Asphaltene Nanoaggregates by DC Conductivity and Centrifugation. *Energy Fuels* **2014**, *28*, 5002−5013.
(35) Goual, L. Impedance Spectroscopy of Petroleum Fluids at Low Frequency. *Energy Fuels* **2009**, *23*, 2090−2094.
(36) Lisitza, N. V.; Freed, D. E.; Sen, P. N.; Song, Y.-Q. Study of Asphaltene Nanoaggregation by Nuclear Magnetic Resonance (NMR). *Energy Fuels* **2009**, *23*, 1189−1193.
(37) Dutta Majumdar, R.; Gerken, M.; Mikula, R.; Hazendonk, P. Validation of the Yen−Mullins Model of Athabasca Oil-Sands Asphaltenes using Solution-State1H NMR Relaxation and 2D HSQC Spectroscopy. *Energy Fuels* **2013**, *27*, 6528−6537.
(38) Eyssautier, J.; Levitz, P.; Espinat, D.; Jestin, J.; Gummel, J.; Grillo, I.; Barre, L. Insight into asphaltene nanoaggregate structure inferred by small angle neutron and X-ray scattering. *J. Phys. Chem. B* **2011**, *115*, 6827−37.
(39) Hoepfner, M. P.; Fogler, H. S. Multiscale scattering investigations of asphaltene cluster breakup, nanoaggregate dissociation, and molecular ordering. *Langmuir* **2013**, *29*, 15423−32.
(40) Eyssautier, J.; Hénaut, I.; Levitz, P.; Espinat, D.; Barré, L. Organization of Asphaltenes in a Vacuum Residue: A Small-Angle X-ray Scattering (SAXS)−Viscosity Approach at High Temperatures. *Energy Fuels* **2012**, *26*, 2696−2704.
(41) Eyssautier, J.; Espinat, D.; Gummel, J.; Levitz, P.; Becerra, M.; Shaw, J.; Barré, L. Mesoscale Organization in a Physically Separated Vacuum Residue: Comparison to Asphaltenes in a Simple Solvent. *Energy Fuels* **2012**, *26*, 2680−2687.
(42) Acevedo, S. c.; Castro, A.; Vásquez, E.; Marcano, F.; Ranaudo, M. a. A. Investigation of Physical Chemistry Properties of Asphaltenes Using Solubility Parameters of Asphaltenes and Their Fractions A1 and A2. *Energy Fuels* **2010**, *24*, 5921−5933.
(43) Mullins, O. C.; Pomerantz, A. E.; Andrews, A. B.; Zuo, J. Y. Asphaltenes Explained for the Nonchemist. *Petrophysics* **2015**, *56*, 266−275.
(44) Barton, A. F. M. *CRC Handbook of Solubility Parameters and Other Cohesion Parameters*, 2nd ed.; CRC Press LLC: United States, 1991.
(45) Fedors, R. F. A method for estimating both the solubility parameters and molar volumes of liquids. *Polym. Eng. Sci.* **1974**, *14*, 147−154.
(46) Hildebrand, J. H.; Prausnitz, J. M.; Scott, R. L. Regular and Related Solutions: The Solubility of Gases. In *Liquids, and Solids*; Van Nostrand Reinhold Company: New York, 1970.
(47) Redelius, P. G. Solubility parameters and bitumen. *Fuel* **2000**, *79*, 27−35.
(48) Lian, H.; Lin, J.-R.; Yen, T. F. Peptization studies of asphaltene and solubility parameter spectra. *Fuel* **1994**, *73*, 423−428.
(49) Frigerio, F.; Molinari, D. A multiscale approach to the simulation of asphaltenes. *Comput. Theor. Chem.* **2011**, *975*, 76−82.
(50) Fossen, M.; Hemmingsen, P. V.; Hannisdal, A.; Sjöblom, J.; Kallevik, H. Solubility Parameters Based on IR and NIR Spectra: I. Correlation to Polar Solutes and Binary Systems. *J. Dispersion Sci. Technol.* **2005**, *26*, 227−241.
(51) Rogel, E.; Ovalles, C.; Moir, M. Asphaltene Stability in Crude Oils and Petroleum Materials by Solubility Profile Analysis. *Energy Fuels* **2010**, *24*, 4369−4374.
(52) Rogel, E.; Ovalles, C.; Moir, M. Asphaltene Chemical Characterization as a Function of Solubility: Effects on Stability and Aggregation. *Energy Fuels* **2012**, *26*, 2655−2662.
(53) Rogel, E.; Ovalles, C.; Vien, J.; Moir, M. Asphaltene Solubility Properties by the In-Line Filtration Method. *Energy Fuels* **2015**, *29*, 6363−6369.

DOI: 10.1021/acs.energyfuels.6b01794
Energy Fuels 2016, 30, 9132−9140

(54) Panuganti, S. R.; Vargas, F. M.; Chapman, W. G. Property Scaling Relations for Nonpolar Hydrocarbons. *Ind. Eng. Chem. Res.* **2013**, *52*, 8009−8020.
(55) Speight, J. G. *Chemical and Physical Studies of Petroleum Asphaltenes*; Elsevier Science B.V., 1994.
(56) Keym, M.; Dieckmann, V. Predicting the Timing and Characteristics of Petroleum Formation Using Tar Mats and Petroleum Asphaltenes: A Case Study from the Northern North Sea. *J. Pet. Geol.* **2006**, *29*, 273−296.
(57) Bunger, J. W.; Cogswell, D. E. Characteristics of Tar Sand Bitumen Asphaltenes as Studied by Conversion of Bitumen by Hydropyrolysis. In *Chemistry of Asphaltenes*; American Chemical Society: Washington, D.C., 1982; Vol. *195*, pp 219−236.
(58) Li, K.; Vasiliu, M.; McAlpin, C. R.; Yang, Y.; Dixon, D. A.; Voorhees, K. J.; Batzle, M.; Liberatore, M. W.; Herring, A. M. Further insights into the structure and chemistry of the Gilsonite asphaltene from a combined theoretical and experimental approach. *Fuel* **2015**, *157*, 16−20.
(59) Behar, F.; Pelet, R.; Roucache, J. Geochemistry of asphaltenes. *Org. Geochem.* **1984**, *6*, 587−595.
(60) Andrews, A. B.; Edwards, J. C.; Pomerantz, A. E.; Mullins, O. C.; Nordlund, D.; Norinaga, K. Comparison of Coal-Derived and Petroleum Asphaltenes by ^{13}C Nuclear Magnetic Resonance, DEPT, and XRS. *Energy Fuels* **2011**, *25*, 3068−3076.
(61) Groenzin, H.; Mullins, O. C. Molecular Size and Structure of Asphaltenes from Various Sources. *Energy Fuels* **2000**, *14*, 677−684.
(62) Pomerantz, A. E.; Le Doan, T. V.; Craddock, P. R.; Bake, K. D.; Kleinberg, R. L.; Burnham, A. K.; Wu, Q.; Zare, R. N.; Brodnik, G.; Lo, W. C. H.; Grayson, M.; Mitra-Kirtley, S.; Bolin, T. B.; Wu, T. Impact of Laboratory-Induced Thermal Maturity on Asphaltene Molecular Structure. *Energy Fuels* **2016**, *30*, 7025−7036.
(63) Sato, T.; Araki, S.; Morimoto, M.; Tanaka, R.; Yamamoto, H. Comparison of Hansen Solubility Parameter of Asphaltenes Extracted from Bitumen Produced in Different Geographical Regions. *Energy Fuels* **2014**, *28*, 891−897.
(64) Painter, P.; Veytsman, B.; Youtcheff, J. Asphaltene Aggregation and Solubility. *Energy Fuels* **2015**, *29*, 2120−2133.
(65) Pomerantz, A. E.; Bake, K. D.; Craddock, P. R.; Kurzenhauser, K. W.; Kodalen, B. G.; Mitra-Kirtley, S.; Bolin, T. Sulfur Speciation in Kerogen and Bitumen from Gas and Oil Shales. *Org. Geochem.* **2014**, *68*, 5−12.
(66) Rogel, E.; Ovalles, C.; Pomerantz, A. E.; Zuo, J. Y.; Mullins, O. C. The Impact of Solubility Parameter Measurements on Predicted Asphaltene Gradients. In *Application to Reservoir Connectivity*; 252nd American Chemical Society National Meeting & Exposition: Philadelphia, PA, 2016.
(67) Maciel, G. E.; Bartuska, V. J.; Miknis, F. P. Characterization of organic material in coal by proton-decoupled ^{13}C nuclear magnetic resonance with magic-angle spinning. *Fuel* **1979**, *58*, 391−394.
(68) Feng, Y.; Le Doan, T. V.; Pomerantz, A. E. The Chemical Composition of Bitumen in Pyrolyzed Green River Oil Shale: Characterization by ^{13}C NMR Spectroscopy. *Energy Fuels* **2013**, *27*, 7314−7323.
(69) Zuo, J. Y.; Mullins, O. C.; Freed, D.; Zhang, D.; Dong, C.; Zeng, H. Analysis of Downhole Asphaltene Gradients in Oil Reservoirs with a New Bimodal Asphaltene Distribution Function. *J. Chem. Eng. Data* **2011**, *56*, 1047−1058.
(70) Zuo, J. Y.; Mullins, O. C.; Mishra, V.; Garcia, G.; Dong, C.; Zhang, D. Asphaltene Grading and Tar Mats in Oil Reservoirs. *Energy Fuels* **2012**, *26*, 1670−1680.
(71) Tissot, B.; Welte, D. *Petroleum Formation and Occurrence*, 2nd ed.; Springer, 1984; p 702.
(72) Okiongbo, K. S.; Aplin, A. C.; Larter, S. R. Changes in Type II Kerogen Density as a Function of Maturity: Evidence from the Kimmeridge Clay Formation. *Energy Fuels* **2005**, *19*, 2495−2499.
(73) Gisolf, A.; Dubost, F. X.; Zuo, J. Y.; Williams, S.; Kristoffersen, J.; Achourov, V.; Bisarah, A.; Mullins, O. C. *Real-Time Integration of Reservoir Modeling and Formation Testing*; Society of Petrophysicists and Well Log Analysts (SPWLA) 50th Annual Logging Symposium: The Woodlands, Texas, 2009; SPWLA RR.
(74) Gisolf, A.; Dubost, F. X.; Zuo, J. Y.; Williams, S.; Kristoffersen, J.; Achourov, V.; Bisarah, A.; Mullins, O. C. *Real Time Integration of Reservoir Modeling and Formation Testing*; SPE EUROPEC/EAGE Annual Conference and Exhibition: Amsterdam, The Netherlands, 2009; SPE 121275.
(75) Dumont, H.; Mullins, O. C.; Zuo, J.; Pomerantz, A. E.; Forsythe, J. C.; Mishra, V. K.; Garcia, G. *Compartments, Connectivity, & Baffling Analyzed by the Extent of Equilibration of Asphaltene Gradients Using DFA*; Offshore Technology Conference: Houston, TX, 2016; OTC 27143.

DOI: 10.1021/acs.energyfuels.6b01794
Energy Fuels 2016, 30, 9132−9140